

Metabolic Thermodynamics and Extended Trophic Cascades in the Web of Life Framework

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Sprint 004 Technical Report

Abstract

Ecosystem simulations frequently compromise physical rigor by treating matter and energy as non-conserved simulation tokens. In Sprint 004, we establish a rigorous thermodynamic foundation for the **Web of Life** simulation engine, grounding all organismal and environmental dynamics in the First and Second Laws of Thermodynamics. By introducing strict interface contracts for internal entropy generation ($\dot{S}_{\text{gen}}$), exergy destruction rate ($\dot{I} = T_0 \dot{S}_{\text{gen}}$), boundary flux arrays, and Redfield-derived stoichiometric mass-balance invariants, we ensure that biological transformations operate as closed-loop thermodynamic systems. Monad-based state pipelines guarantee immutable stock transitions, eliminating side-effect leakage while preserving strict mass conservation ($\pm 10^{-9}$ g) and unidirectional thermodynamic dissipation ($dQ_{\text{loss}}/dt \geq 0$).

1 Introduction and Thermodynamic Formulation

The transition from token-based simulation models to physically bounded ecological systems requires strict adherence to physical conservation laws. Sprint 004 implements a thermodynamic state vector interface (`src/thermodynamics/types.ts`) that governs all metabolic processes within the ecosystem.

1.1 First Law Compliance (Matter Conservation)

For any biological subsystem or organism S , the temporal evolution of mass is governed by elemental stoichiometry (Carbon, Nitrogen, Phosphorus, Oxygen, and Hydrogen):

$$\frac{dM_S}{dt} = \sum \text{Ingestion} - \sum \text{Excretion} - \sum \text{Respiration}_m - \text{Mortality} \quad (1)$$

Atomic mass conservation is rigorously enforced across all biochemical transformations, including photosynthesis, cellular respiration, herbivory, and saprotrophic mineralization.

1.2 Second Law Compliance (Entropy & Dissipation)

The system receives external energy solely via Photosynthetically Active Radiation (Q_{in}). All internal metabolic processes generate unrecoverable thermal dissipation (Q_{loss}), driving ecosystem entropy upward:

$$\frac{dQ_{\text{loss}}}{dt} \geq 0 \quad (2)$$

Ecological efficiency η is bounded by thermodynamic limits ($\eta \leq 0.25$), satisfying Lindeman's trophic efficiency constraints.

2 Mathematical Model and Stoichiometric Conversions

All entities operate on Redfield-derived terrestrial biological ratios scaled for atomic mass units (AMU):

- **Stoichiometric Ratio (C : N : P):** 106 : 16 : 1 (molar), translated to 45% C, 4% N, 0.5% P by dry weight.
- **Photosynthesis:** $6CO_2 + 6H_2O + \text{PAR (2870 kJ/mol)} \rightarrow C_6H_{12}O_6 + 6O_2$
- **Respiration:** $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{Energy (2870 kJ/mol)} + Q_{loss}$

3 Monad State Pipelines and Verification

State updates are executed through composable Monad pipelines (**ThermodynamicMonad**), validating invariant conditions at every simulation tick:

1. **Global Mass-Balance Invariant:** $\sum M_{\text{Environment}} + \sum M_{\text{Organisms}} + \sum M_{\text{Detritus}} = \text{Constant} \pm 10^{-9} \text{ g.}$
2. **Second Law Monotonicity:** $\Delta Q_{\text{loss}} \geq 0$ for every agent update step.
3. **Lindeman 10% Rule:** Steady-state biomass production ratios across trophic levels conform to $0.09 \leq \frac{B_{n+1}}{B_n} \leq 0.12$.

4 Conclusion

Sprint 004 successfully transitions the **Web of Life** platform into a thermodynamically rigorous ecosystem simulator. All developments are publicly archived in the official repository: <https://github.com/pascalranoroarijaona/WebOfLife>.